

Abdul sami *Full Stack Software Engineer*

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SUMMARY

Full Stack AI Engineer with 5+ years of experience engineering high-performance web applications, specialized in combining the **MERN/Laravel** stack with advanced **Generative AI, RAG architectures, and multi-agent workflows**. Expert in constructing resilient data ingestion pipelines, integrating Large Language Models (LLMs), orchestrating tool/function-calling agents, and deploying vector search frameworks. Adept at managing the entire AI lifecycle—from prompt optimization and context window budget management to robust production delivery within remote-first environments.

SKILLS

AI Stack & Orchestration: Generative AI, Retrieval-Augmented Generation (RAG), AI Agents, Function Calling, Prompt Engineering, LangChain / LlamaIndex / LangGraph, AI Observability & Evals. | **Vector Databases & Data:** Milvus, Neo4j, pgvector, Semantic Search, Chunking Strategies, Knowledge Graphs. | **Back-End & APIs:** Node.js, Express.js, Python, PHP, Laravel, RESTful APIs, WebSockets (Socket.io [🔗](#)), Microservices. | **Front-End Development:** React.js, Redux, Vue.js, TypeScript, Context API, Tailwind CSS, Responsive UI. | **DevOps & Cloud Infrastructure:** AWS, Docker, Ubuntu VPS, Nginx (Reverse Proxy & Subdomain Routing), CI/CD Pipelines, Git/GitHub/GitLab, Linux Server Management | **Testing & Architecture::** Postman, API Testing, PHPUnit, Jest, Distributed Systems (RabbitMQ, Socket.io [🔗](#)), Performance Optimization, Jira, Redis

PROFESSIONAL EXPERIENCE

Syanax PTV Limited, Software and Ai Engineer

04/2024 – Present
Lahore, Pakistan

- **Architected and productionized intelligent, agentic systems** and real-time operations dashboards using Node.js, React.js, and Python, streamlining complex workforce management workflows.
- **Engineered an advanced RAG pipeline** using an orchestration layer (LangChain/LlamaIndex) to index, parse, and dynamically query large-scale operations manuals, significantly boosting contextual data retrieval accuracy.
- **Developed autonomous AI agents with function-calling capabilities** to trigger automated workflows, execute backend tasks, and parse live streaming data (weather, task logs) via Node.js microservices.
- **Implemented vector semantic search** utilizing optimized database engines (Redis/Milvus) for rapid, low-latency document chunk retrieval and token budget management.
- **Optimized real-time server communication** with Socket.io [🔗](#) to stream LLM responses token-by-token directly to frontend Vue/React views, decreasing perceived UI latency.
- **Designed and secured high-throughput RESTful APIs** utilizing JWT and strict role-based access management to shield exposed AI service layer endpoints.

IPLEX, Senior PHP Laravel Developer

10/2023 – 03/2024
Islamabad, Pakistan

- **Spearheaded full stack development** using React.js, Node.js, Express.js, and MongoDB to deliver high-priority client-facing web applications and internal tooling automation.
- **Built reusable React component libraries** and dynamic analytics dashboards that successfully visualized data-heavy backend logs and complex business workflows.
- **Enhanced backend responsiveness** by refactoring Express.js middleware structures and optimizing MongoDB aggregation pipelines for efficient data mutations.

Rapidev.tech, PHP Laravel Developer

05/2021 – 10/2023
Sargodha, Pakistan

- **Engineered full stack applications** across MERN-stack backend architectures, supporting core operations behind healthcare, payment processing, and hybrid mobile applications.
- **Integrated third-party APIs**, secure payment gateway workflows (Stripe), and automated transaction reporting mechanisms across distributed multi-tenant services.
- **Refactored legacy database-heavy logic**, optimizing complex SQL/NoSQL schemas to isolate long-running operations and prevent runtime transaction deadlocks.

PROJECTS

ASB taskTracker, A large-scale golf course management and workforce tracking platform designed to manage daily operations, employee activities, weather forecasting, mowing schedules and task monitoring.

- Developed and maintained full-stack modules using Laravel and Vue.js and automated it using Ai.
- Implemented real-time communication features using Socket.io [🔗](#)
- Integrated Mapbox for location-based task visualization and tracking.
- Worked with Redis and RabbitMQ for queue management and scalable background processing.

- Collaborated within a microservices-based architecture to improve system performance and maintainability.
- Contributed to modernization of a legacy Core PHP system into a scalable Laravel + Vue.js platform.

ASB Golf Operations Intelligent System & Agentic Workboard, *An enterprise-grade golf operations management platform featuring real-time employee tracking, weather intelligence, and automated task allocation.*

- **Designed and implemented an AI Agent workflow** using Python and Node.js that dynamically assigns and schedules golf course tasks based on real-time external data streams (weather variations, employee skill tracking, mowing rules).
- **Integrated a dedicated RAG system** backed by a Vector Database to enable natural-language interrogation of equipment manuals and regulatory operations records.
- **Leveraged Mapbox API and Socket.io**  to provide immediate, real-time geolocated UI updates corresponding directly to agentic adjustments on the interactive operations board.

Golf Yield AI: Dynamic Pricing Model,

An end-to-end machine learning solution designed to maximize tee-sheet revenue for golf operations.

- Developed and trained a predictive regression model using **CatBoost** to forecast optimal tee time pricing based on real-time weather analytics, historical occupancy trends, and booking lead times.
- Deployed the model as an independent, lightweight microservice accessible via REST APIs to dynamically inject pricing shifts into the booking engine.
- **Technologies:** Python, CatBoost, FastAPI, Pandas, NumPy

Syanex Official Website,

The official enterprise web portal built with a highly responsive frontend and a secure, hardened backend infrastructure.

- Engineered secure authentication modules and API protection layers using Laravel Sanctum.
- Integrated automated communication layers via SendGrid and spam-prevention mechanisms using Google reCAPTCHA.
- **Technologies:** Laravel, Vue.js, Bootstrap, MySQL, Sanctum, SendGrid, REST APIs

Payment CMS, *An organizational transaction management and billing engine developed for enterprise payment services.*

- Integrated Stripe Payment APIs, establishing secure 3D Secure authentication workflows, refund handling, and affiliate payouts.
- Built advanced transaction tracking and comprehensive financial reporting modules.
- **Technologies:** Laravel, PHP, Stripe API, MySQL, Bootstrap, jQuery

EDUCATION

Bachelor of Software Engineering, *University of Sargodha (UOS) | 08/2019 - 07/2023*

Core Focus Areas: Artificial Intelligence, Cloud Computing, Data Structures & Algorithms, Database Systems, Web & Mobile Application Development, Object-Oriented Programming